

Cover Letter for LIQUIDICE project, Position 2, “machine learning regional climate emulator to be able to complement the dynamical downscaling methods for high resolution climate projections”.

Dear ICTP recruitment office, I'm Agostino Manzato from Italy and my main interest is in climatology and forecasting of convective events (heavy rainfall, thunderstorms, hail, lighting strikes...). For that reason I developed specific skills for forecasting these events at regional scale, from observations or model data output, using automatic tools like machine learning methods (NN and, more recently, Convolutional NN).

I do not have a PhD, but if you look at my CV and publication record (see <https://scholar.google.com/citations?hl=it&user=7-6xnYEAAAAJ> or <https://orcid.org/0000-0001-8339-3062>) you will see that I have gained a lot of experience, including in forecast verification field, thanks to the many conferences, schools and workshops (including at ICTP), that I was able to follow in the last 30 years. In particular, in 2023 I successfully completed the Machine Learning in Weather & Climate course (MOOC up to 40 hours) organized online by ECMWF and IFAB.

I also was tutoring students at the Udine University. In particular, in 2018 and 2019 I was tutor of two students of the Department of Mathematics, Computer Science and Physics during their thesis on thunderstorm forecasting with convolutional neural networks (CNN) and analogue methods forced by genetic algorithms (AtmoSwing). See https://presentations.copernicus.org/ECSS2019/ECSS2019-222_presentation.pdf

I work since July 2000 for ARPA FVG, but in the last period I took one year of leave to work for ISAC-CNR, Bologna, but mainly working from the Udine University headquarter (SAWEB informatic Lab) in Gorizia, where I live.

Best regards!

Gorizia, 28 May 2025

A handwritten signature in black ink, appearing to read 'Agostino Manzato', written over a horizontal line.